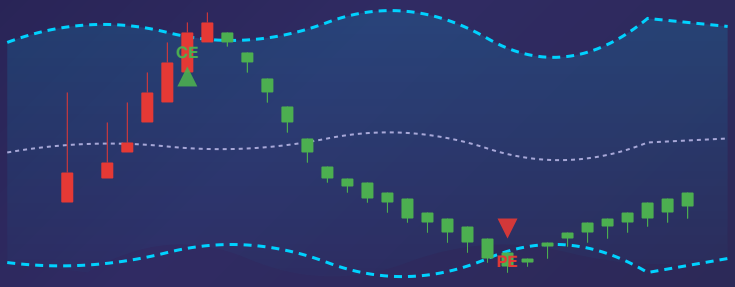
NIFTY OPTIONS

Scalping Strategy

SCALP_BB_RSI_V4 — 3-Minute Bollinger Band + RSI Scalping



Version 4.0 • 3-Min Timeframe • BB + RSI + PSAR

Automated Entry • PSAR Trailing SL • Profit Locking • VIX Filter

1. Strategy Overview

The **SCALP_BB_RSI_V4** strategy is a **3-minute** timeframe scalping system designed for **NIFTY index options** (CE/PE). It identifies short-term momentum breakouts using Bollinger Bands and RSI, with dynamic stop-loss management via Parabolic SAR.

3 min	BB+RSI	PSAR	50 pts
Candle Timeframe	Entry Signal	Trailing SL	Max SL Cap

2. Indicators Used

Bollinger Bands (BB)

Measures volatility envelope around price. Entry triggered when price breaks above upper band (CE) or below lower band (PE).

Period: 20 | StdDev: 1.0

RSI (Relative Strength Index)

Momentum confirmation filter. Ensures price breakout has directional strength before entering.

Period: 14 | CE > 55 | PE < 45

Parabolic SAR (PSAR)

Dynamic trailing stop-loss. Tightens every candle. Also used for initial SL calculation and reversal detection (SAR flip).

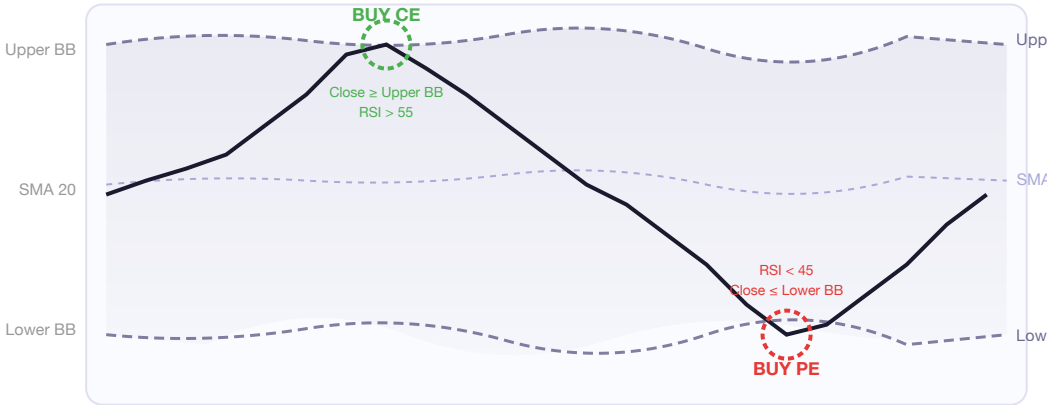
Step: 0.02 | Max: 0.2

VIX Filter (Market Regime)

Blocks entries when India VIX is too high, avoiding choppy/whipsaw conditions that hurt scalping performance.

Max: 20 | Strong Only: 16

3. How BB + RSI Signal Works



4. Entry Logic

BUY CE (Bullish Entry)

1. Candle close≥ **Bollinger Upper Band**(price breaking above volatility envelope)
2. RSI >**55**(momentum confirmation — not just a wick)
3. Stop Loss =**tighter of**(Previous candle LOW, PSAR value), capped at 50 pts
4. Select**ATM Call option**based on current NIFTY spot price

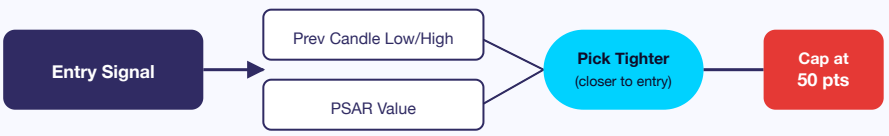
BUY PE (Bearish Entry)

1. Candle close≤ **Bollinger Lower Band**(price breaking below volatility envelope)
2. RSI <**45**(bearish momentum confirmation)
3. Stop Loss =**tighter of**(Previous candle HIGH, PSAR value), capped at 50 pts
4. Select**ATM Put option**based on current NIFTY spot price

Pre-Entry Validations

Check	Condition	Action if Failed
Market Hours	9:21 AM – 2:30 PM IST	Skip entry
Candle Warmup	≥ 30 candles loaded	Wait for more data
Daily Loss Limit	Day P&L > -₹2,000	Block all entries for day
Max Daily Trades	< 30 trades today	Block further entries
SL Pause Cooldown	2 candles since last SL hit	Wait for cooldown
VIX Filter	India VIX ≤ 20	Block entry (see VIX section)
No Open Position	Not already in a trade	Skip signal

5. Stop Loss Calculation



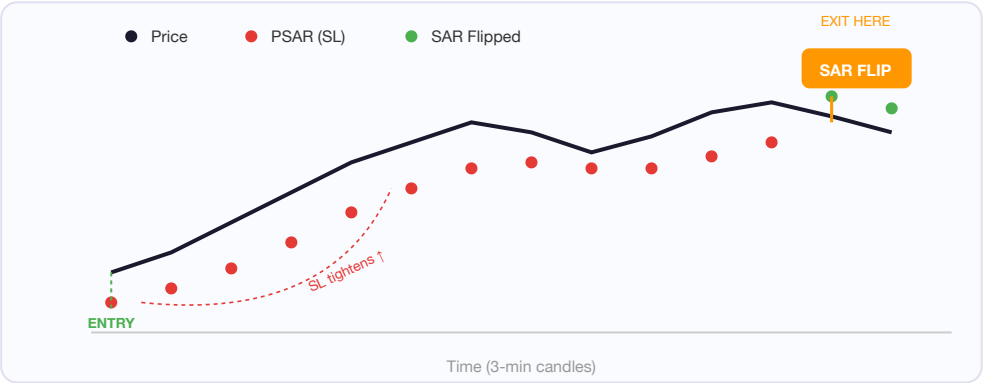
Example (CE Entry at 22,500):
Prev Candle Low = 22,465 → SL distance = 35 pts
PSAR = 22,470 → SL distance = 30 pts → Pick PSAR (tighter) → Final SL = 22,470 (30 pts)

6. Exit Logic

Exits are evaluated on every candle in the following priority order:

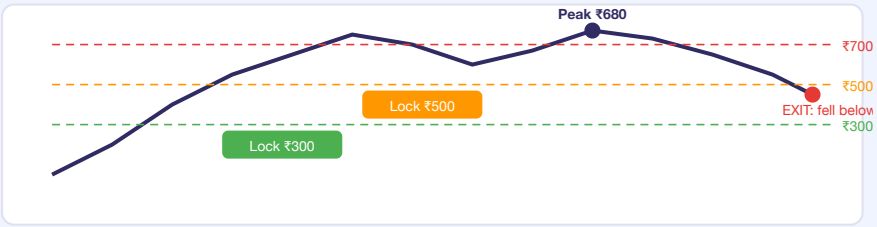
Priority	Exit Type	Trigger Condition	Details
P1	**PSAR Stop Loss**	Price crosses SL level	Initial SL from entry; tightens every candle (never widens). CE: SL moves up. PE: SL moves down.
P2	**Trailing Profit Lock**	PNL drops below locked level	Activates at ₹300 peak profit. Locks in steps of ₹200 (300, 500, 700, 900...)
P3	**SAR Flip (Reversal)**	PSAR crosses to opposite side	Indicates trend reversal. Immediate exit regardless of P&L.
P4	**EOD Square-off**	Time = 3:20 PM IST	All open positions closed. No new entries after 2:30 PM.

PSAR Trailing Stop Loss Visualization



Trailing Profit Lock Mechanism

How it works: Once peak unrealized P&L crosses ₹300, profit locking activates. It locks in steps of ₹200.



Example: Peak profit reaches ₹680 → locked at ₹500 (highest complete step). When P&L drops to ₹490 → EXIT triggered, protecting ₹500 profit.

7. VIX Market Regime Filter

The India VIX (volatility index) determines whether market conditions suit scalping. High VIX causes whipsaws that repeatedly trigger stop losses.

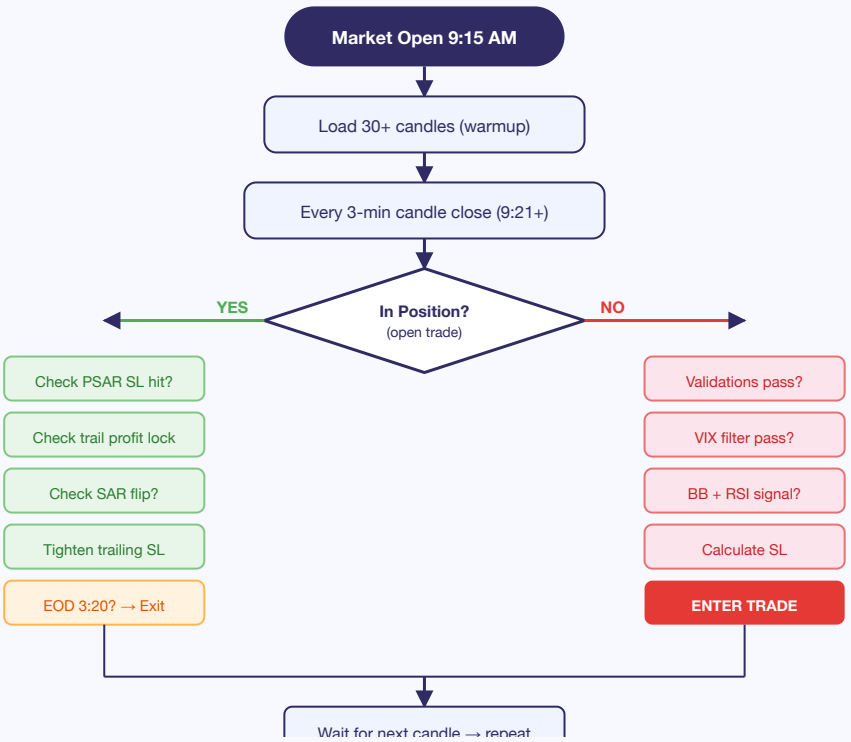
VIX ≤ 16	16 < VIX ≤ 20	VIX > 20
ALL signals allowed	STRONG signals only	ALL entries BLOCKED
Calm market, ideal for scalping	Elevated volatility, cautious	High volatility, too risky

8. Risk Management

₹2,000	30	2	50 pts
Max Daily Loss	Max Trades/Day	SL Pause Candles	Hard SL Cap

SL Pause: After a stop loss is hit, the strategy pauses for 2 candles (6 minutes) to avoid re-entering immediately into the same choppy move.

9. Complete Trade Flow



Confidential

10. Configuration Parameters

All parameters are configurable via `.env` file or the Settings UI:

Core Settings

Parameter	Default	Description
SCALP_ENABLED	false	Enable/disable scalp strategy
SCALP_RESOLUTION	3	Candle timeframe in minutes (1/2/3/5)
SCALP_ENTRY_START	09:21	Entry window start time (IST)
SCALP_ENTRY_END	14:30	Entry window end time (IST)

Indicator Settings

Parameter	Default	Description
SCALP_BB_PERIOD	20	Bollinger Bands lookback period
SCALP_BB_STDDEV	1.0	BB standard deviation multiplier
SCALP_RSI_PERIOD	14	RSI calculation period
SCALP_RSI_CE_THRESHOLD	55	RSI threshold for CE entry
SCALP_RSI_PE_THRESHOLD	45	RSI threshold for PE entry
SCALP_PSAR_STEP	0.02	PSAR acceleration step
SCALP_PSAR_MAX	0.2	PSAR max acceleration
SCALP_MAX_SL_PTS	50	Hard cap on stop loss distance (points)

Profit & Risk Settings

Parameter	Default	Description
SCALP_TRAIL_START	300	Trailing profit activation threshold (₹)
SCALP_TRAIL_STEP	200	Trailing profit lock step size (₹)
SCALP_MAX_DAILY_TRADES	30	Maximum trades per day
SCALP_MAX_DAILY_LOSS	2000	Daily loss limit (₹)
SCALP_SL_PAUSE_CANDLES	2	Cooldown candles after SL hit

VIX Filter Settings

Parameter	Default	Description
VIX_FILTER_ENABLED	true	Enable VIX-based market regime filter
VIX_MAX_ENTRY	20	Block ALL entries above this VIX level
VIX_STRONG_ONLY	16	Above this: only strong signals allowed

11. Operating Modes

Live Trading

Paper Trading

Real orders via Fyers API. Uses WebSocket for real-time ticks. Builds 3-min candles from tick data. 1-second option LTP polling.

scalp.js

Simulated fills with same WebSocket data and exit logic. No real orders placed. Identical strategy evaluation.

scalpPaper.js

Backtesting

Historical data from Fyers API. Simulates option pricing with delta (0.55) and theta (10 pts/day). CPR and VIX filters.

scalpBacktest.js

Settings UI

All thresholds configurable via web-based Settings page. Changes apply immediately without restart.

Dynamic Config

Key Design Principles

No Fixed Targets: All exits are dynamic — SL hits, trailing locks, SAR reversals, or EOD. The strategy lets winners run.

Tightening-Only SL: Stop loss can only move closer to price, never relax. This protects gains but may exit early in choppy moves.

Multi-Layer Safety: VIX filter + daily loss limit + max trades + SL pause = multiple defensive layers preventing excessive losses.